

THOMAS POINT, MD, USA

WMO: 994400

Lat: 38.899N

Lon: 76.436W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.4	(c) -6.1	(d) -19.8	(e) 0.6	(f) -5.7	(g) -17.1	(h) 0.8	(i) -4.7	(j) 18.0	(k) 6.6	(l) 15.9	(m) 3.2	(n) 7.7	(o) 310	(p) 0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 4.9	(c) 30.5	(d) 24.7	(e) 29.4	(f) 24.4	(g) 28.5	(h) 24.0	(i) 27.7	(j) 28.6	(k) 26.6	(l) 27.8	(m) 25.6	(n) 27.1	(o) 4.8	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.5	23.3	28.1	26.3	21.7	27.2	25.3	20.4	26.4	87.9	28.5	82.9	27.7	78.8	27.0	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 17.0	(b) 14.3	(c) 11.9	(e) -10.3	(f) 33.0	(g) 3.3	(h) 1.4	(i) -12.6	(j) 34.0	(k) -14.6	(l) 34.8	(m) -16.4	(n) 35.6	(o) -18.8	(p) 36.7		
(a) 17.0	(b) 14.3	(c) 11.9	(e) -11.8	(f) 28.5	(g) 2.2	(h) 1.9	(i) -13.4	(j) 29.9	(k) -14.7	(l) 31.0	(m) -15.9	(n) 32.1	(o) -17.5	(p) 33.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.9	2.0	3.0	6.8	12.8	17.9	22.7	25.3	24.8	21.5	15.5	9.7	4.8
	DBStd	9.01	5.00	4.36	4.26	3.84	3.48	2.93	2.16	2.11	2.80	3.70	4.02	4.22
	HDD10.0	809	252	199	118	15	0	0	0	0	0	4	53	168
	HDD18.3	2307	508	430	358	171	52	4	0	0	6	102	258	419
	CDD10.0	2250	3	3	18	98	245	379	473	458	344	175	45	8
	CDD18.3	706	0	0	0	4	38	134	215	199	100	15	0	0
	CDH23.3	4314	0	0	2	16	106	687	1687	1412	378	27	0	0
	CDH26.7	769	0	0	0	1	12	103	371	249	29	4	0	0
Wind	WSAvg	5.8	6.1	6.1	6.6	6.5	5.8	5.3	5.0	5.0	5.6	6.0	6.2	5.9
Precipitation	PrecAvg	1168	89	67	108	92	94	107	113	114	109	98	80	94
	PrecMax	1673	176	195	235	167	209	322	215	207	312	209	166	208
	PrecMin	896	35	7	0	39	19	56	40	30	14	3	14	35
	PrecStd	176	34	42	48	31	51	60	50	47	69	62	47	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.6	17.5	21.3	25.5	28.4	30.9	32.9	31.8	29.3	25.8	20.4	18.1
		MCWB	12.7	11.9	15.8	16.8	23.2	23.5	25.9	23.5	23.3	22.2	16.2	14.7
	2%	DB	12.6	12.8	17.4	22.5	25.8	29.1	30.9	30.1	27.4	23.7	18.0	14.8
		MCWB	10.0	8.7	13.3	16.3	21.7	23.5	25.6	24.1	23.5	20.9	15.2	12.9
	5%	DB	10.5	10.3	14.8	20.1	24.3	27.9	29.8	29.0	26.4	22.3	16.6	12.4
		MCWB	8.7	7.3	11.7	15.6	21.2	23.3	25.1	23.9	22.9	19.7	14.3	10.9
	10%	DB	8.5	8.4	12.7	18.2	22.9	26.7	28.7	28.0	25.5	20.8	15.3	10.3
		MCWB	6.4	6.0	10.1	14.5	20.2	23.0	24.6	23.5	22.0	18.3	13.1	8.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	13.4	17.2	20.5	26.3	27.9	29.9	27.6	26.8	23.5	18.2	15.4
		MCDB	16.0	16.0	19.1	21.6	27.2	28.6	30.8	28.4	27.2	24.1	18.9	16.1
	2%	WB	11.2	9.7	14.7	18.3	23.9	26.3	28.4	26.5	25.3	22.3	16.2	12.8
		MCDB	12.6	11.3	16.5	20.7	24.7	27.0	29.1	27.9	26.2	23.7	17.3	13.9
	5%	WB	9.1	7.5	12.6	17.0	22.4	25.1	27.3	25.7	24.4	20.9	15.0	10.9
		MCDB	10.3	9.1	14.1	19.6	23.7	26.4	28.4	27.3	25.7	22.1	15.9	11.8
	10%	WB	6.8	5.7	10.7	15.7	21.3	24.2	26.1	25.0	23.5	19.3	13.6	9.0
		MCDB	8.0	7.3	12.0	18.0	22.8	25.9	27.7	26.8	25.0	20.7	14.6	9.8
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.7	6.2	6.7	5.8	5.1	4.9	4.7	4.8	5.2	5.2	4.9
		MCDBR	8.3	8.8	9.5	9.5	7.1	6.2	5.9	5.5	5.1	6.1	6.6	7.6
		MCWBR	7.1	7.1	6.7	5.2	4.4	4.4	4.4	4.0	4.1	4.8	5.7	7.2
	5% WB	MCDBR	8.6	8.0	9.1	8.3	5.8	5.4	5.1	4.8	4.5	5.4	5.6	7.0
		MCWBR	7.7	7.1	7.6	5.8	4.1	4.4	4.6	4.2	4.1	4.9	5.7	7.3
Clear-Sky Solar Irradiance	taub	0.312	0.320	0.359	0.398	0.442	0.494	0.518	0.511	0.421	0.362	0.326	0.311	
	taud	2.484	2.446	2.394	2.304	2.214	2.114	2.062	2.067	2.291	2.461	2.519	2.526	
	Ebn at Noon	865	905	898	879	843	797	773	767	822	846	842	841	
	Edh at Noon	79	93	109	127	142	157	164	159	120	91	76	71	
All-Sky Solar Radiation	RadAvg	2.10	2.95	3.92	4.99	5.63	6.23	6.12	5.44	4.42	3.30	2.39	1.85	
	RadStd	0.17	0.33	0.31	0.38	0.67	0.47	0.29	0.31	0.49	0.36	0.20	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+120	N/A
(47) Regional (81 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	+82

Nomenclature: See separate page